

## **Question & Answer**

**Q.** For packet 1: What kind of HTTP packet is packet no. 1?

**Answer:** HTTP Request

**Q.** Click onto “Inbound PDU details” tab. Scroll down at the end, what do you see?

**Answer:** We can see HTTP request details at the end of the Inbound PDU details, the website I want to host or render “Host: [www.bracu.ac.bd](http://www.bracu.ac.bd)”.

“HTTP Data:Accept-Language: en-us

Accept: \*/\*

Connection: close

Host: www.bracu.ac.bd”

**Q.** For packet 2: Click onto “Inbound PDU details” tab. Scroll down at the end, what do you see? What kind of HTTP packet is this?

**Answer :** We can see HTTP response details at the end of the Inbound PDU details.

“HTTP Data:Connection: close

Content-Length: 151

Content-Type: text/html

Server: PT-Server/5.2”

And the packet type is HTTP Response.

**Q.** For packet 4: What is the purpose of this DNS packet?

**Answer:** This DNS packet is requesting the IP address for the domain name "[mail.bracu.ac.bd](mailto:mail.bracu.ac.bd)" (BRAC University's mail server). It asks the local DNS server to translate "[mail.bracu.ac.bd](mailto:mail.bracu.ac.bd)" into its corresponding IP address so the device can establish a connection.

**Q.** For packet 5& 6: Explain why SMTP packet was sent to the email server and the server replied with an SMTP packet?

**Answer:**

**Packet 5 (PC1 → Email Server):** The SMTP packet was sent to transfer the outgoing email from the client to the mail server.

**Packet 6 (Email Server → PC1):** The server replied with an SMTP acknowledgment packet to confirm successful receipt of the email.

**Q.** For packet 6: What is the purpose of this DNS packet?

**Answer:** The purpose of this DNS packet is to resolve the domain name of the email server into its corresponding IP address. This is necessary before the POP3 client can establish a connection with the email server to retrieve emails.

**Q.** For packet 7&8: Explain why POP packet was sent to the email server and the server replied with a POP packet?

**Answer:**

**Packet 7 (PC0 → Email Server):** The POP3 packet was sent from the client to the email server to request and retrieve incoming emails from the mailbox. Here, POP3 is connected to PC1.

**Packet 8 (Email Server → PC0):** The server replied with a POP3 packet to deliver the requested emails. The POP3 protocol uses a request-response mechanism where the server sends acknowledgments and the actual email data back to the client, confirming the retrieval operation.